Composed inverse functions in the network search

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Goal. Continue the working norm-network loop with the actual norm–inverse–product–cleanup dependencies. Reuse the existing reverse-jet callsite map and883-qubit incumbent. First discriminate whether exact composed Boolean functions expose useful sharing that separate quadratic blocks hide. Then lower admissible shared-function alternatives, verify their phase/cleanup, and screen them using the existing global live-register envelope. No gain is promised before measurement.

Ownership. Core retains the authored typed function/operation model. Graph supplies exact binary bases and reconstruction witnesses through its existing example/facade; no consumer rank solver. Quantum owns bounded Boolean coefficient evaluation, circuit lifetime lowering and phase contracts. Pgraph MSG/SSG remains the network generator. The native adapter and HermiteLibrary own binding to unchanged source-bound public runtime and native field verification.

Discriminating experiment. At six input bits, enumerate all64 values and exact ANF coefficients of input bits, norm products, inverse-norm bits and proposed products. Compare prefix and combined ranks and reconstruct the requested inverse/zero flag. A basis change alone cannot lower a full-rank space. A useful relation must remove an actual paid product after prefix costs, cleanup and retained qubits are counted. Keep alternative prefixes and resulting networks; no isolated long random-flip campaign.

Implementation scope. Extend arithmetic.py/source vocabulary and Quantum synthesis only where the experiment supports a valid lowering. Reuse quadratic factor emission, exact basis verification and existing network feedback. If explicit column weights are needed, extend only the Graph CLI adapter to its existing binary-weighted-basis API, with source-bound costs; no provider solver rewrite. Files: arithmetic.py, reversible.py, hermite.py as needed; synthesis/arithmetic.rs, network.rs and a cohesive composed-function module if required; source metadata/fixture, CLI, focused tests, quantum_network.py, existing report.py and README. Exact final list belongs in the report.

Verification. Algebra certificates compare all64 inputs and every reconstructed function, including zero. Small unitary XOR paths check all4096 input/target states. Fresh-output/measured cleanup must have an explicit separate contract and phase proof. Native runs serialize/reload circuits and use independent GF arithmetic, then cross-pair affected jet directions. Only an equivalent, globally promising candidate receives the102400-shot official local verifier. No local-block Q-squared-T improvement is reported as a whole-decoder result.

Caps and resources. Preserve existing24-wire/12-data/512-gate limits on the bounded primitive verifier; larger native jets retain their separate limits. At most1600 proposed product columns,64 coefficient coordinates and 16 observed networks initially; no unbounded SSG. Native probe guard900s, one CPU/2GiB for small kernels,12GiB/3600s only if full verification is justified. Amaterasu has313GiB free and about19GiB available RAM. Expected60–100min for model/certificate work, tests and native comparison; resource monitoring is a guard, not an unsolicited speed benchmark. No GPU, WSL or source-tree copies.

Repository and risk. Transformation feat/unversioned c90eb2b, fresh fetch succeeds and upstream is current. Preserve unrelated changes. Graph source is read-only unless the CLI weight adapter proves necessary;

fetch its remotes before edits. Existing uncommitted Graph API remains a recorded local integration dependency. No new dependency or CORE.YAML-protected item is planned. No commit/push/submission. Rollback only named task edits, keep prior kernels. Worst case: composed rank gives no sharing, or retained cache loses at global qubit peak; report that measured limit and do not promote the candidate.

Refinement,02:27. The separated input-times-inverse family has rank35 including a16-dimensional free prefix, so its full product span does not contract. Include mixed prefix parities to test identities involving cached norm products; at most57 operand forms give1596 pairs. This motivates the1600 column cap above before the mixed-family implementation.

Refinement,02:34. Minimizing a basis for the whole supplied span does not minimize reconstruction of the seven requested outputs. Extend Graph's binary matroid owner with bounded exact basis exchanges and a required-column beam objective, retaining the already available prefix. This is a provider API, not a consumer rank solver. Add direct reconstruction/certificate tests; cap exchange evaluations at4096 and retain four states. The native gate search will consume the resulting certificates through the same Core source identity.

